2520090063

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OSSP:

Commands:

uname : Displays basic operating system information.

lscpu : Displays CPU and processor details.

```
Radhakrishna@04:~# nano Week-2_T1.c
Radhakrishna@04:~# gcc Week-2_T1.c -o Week-2_T1
Radhakrishna@04:~# ./Week-2_T1
Enter a Linux command: uname

Parent Process
Parent PID : 520
Child PID  : 521

Child Process
Child PID  : 521
Parent PID : 520

Executing command: uname

Linux

Child process completed.
Radhakrishna@04:~#
```

ps : Displays the currently running processes.

```
Radhakrishna@04:~#  nano Week-2_T1.c
Radhakrishna@04:~# gcc Week-2_T1.c -o Week-2_T1
Radhakrishna@04:~# ./Week-2_T1
Enter a Linux command: ps

Parent Process
Parent PID : 538
Child PID  : 539

Child Process
Child PID  : 539
Parent PID : 538

Executing command: ps

    PID TTY          TIME CMD
    305 pts/0    00:00:00 bash
    538 pts/0    00:00:00 Week-2_T1
    539 pts/0    00:00:00 ps

Child process completed.
Radhakrishna@04:~#
```

Relation Between Hardware Resources and Operating System Services:

1. CPU

The operating system manages the CPU by dividing its processing time among multiple applications.

Example**:** When we open Chrome and a music player together, the OS schedules CPU time for both.It Process Both Chrome And music player

2. Memory (RAM) -> Memory Management

The operating system assigns a separate portion of RAM to every running application.

Example: Chrome and VS Code each get their own memory space while running.

3. Storage (SSD) → File System Management

The operating system storesand organizes files on the storage device.
Example: Saving a document in the Documents folder on our computer.

4. I/O Devices → Device Management

The operating system uses device drivers to control input and output devices.